

A Single-Particle Tracking Framework for Quantifying Intracellular Partition Free Energies

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ABSTRACT Biomolecular condensates are membrane-less organelles that locally enrich specific molecules at spatial sites. A central challenge is determining the physical parameters of individual condensates, as their dimensions are typically below the diffraction limit of conventional microscopy. Single-particle tracking overcomes this limitation, but robust methods for extracting physical observables from trajectory collections remain scarce. Here, we present a framework for inferring the potential difference between molecules inside a condensate and those in the surrounding environment. We introduce a correction that accounts for condensate motion during acquisition, enabling reliable density estimation even from sparse data. After testing on *in silico* data, we apply our approach to 5 experimental datasets of proteins active in DNA repair and separate them based on the liquid nature of the focus.

SIGNIFICANCE Quantifying molecular motion, forces, and free-energy landscapes in crowded cellular environments remains a major challenge in both biological physics and soft-matter research. We present a computational framework that directly infers diffusion coefficients and potential energy profiles from single-particle tracking trajectories. To address condensate motion during imaging, we implement a center-correction procedure that significantly improves the robustness of energy measurements. By implementing spatial Monte Carlo sampling, our method reliably estimates free energies even from sparse datasets, making it applicable to a wide range of existing datasets. Together, these tools enable model-driven classification of condensates and provide a broadly applicable approach for investigating intracellular compartmentalization across biological and soft-matter systems.

INTRODUCTION

Cells rely on specialized proteins that must locate specific targets within the intracellular environment to perform their functions (1). Because macromolecules occupy approximately 30–40% of the nuclear volume, diffusion is restricted and long-range search becomes inefficient (2). To overcome this challenge, cells have evolved membrane-less sub-compartments, commonly referred to as biomolecular condensates or foci (hereafter referred to as focus/foci). These compartments locally increase the concentration of specific proteins, reduce interference between unrelated processes and accelerate reaction rates.

Foci have been observed for decades (3–10), yet the physical principles that govern their formation remain an active area of research. Two principal models have been proposed to explain foci formation (11): 1) *Polymer–Polymer Phase Separation (PPPS)*, in which strong interactions between proteins and chromatin establish dense clusters of binding sites that bridge distant genomic loci. 2) *Liquid–Liquid Phase Separation (LLPS)*, where weak, multivalent interactions between proteins form dynamic, liquid-like droplets maintained through reversible interactions.

A classical example is the focus formed around a DNA double-strand break (DSB), where the concentration of repair proteins increases to ensure efficient genome maintenance (12). The DSB focus has been proposed to form through LLPS, supported by experimental observations ranging from Single-Particle Tracking (SPT) to wide-field microscopy (7, 13–15). A central open question is whether this focus remains liquid under physiological conditions, or transitions to a more solid-like

state in pathological scenarios, potentially revealing weaknesses of the DNA damage response. More broadly, understanding the physical mechanisms behind the many foci formed inside cells remains a crucial challenge in molecular biophysics (11, 16–18).

A key obstacle to studying focus properties is that the spatial and temporal resolution required to extract meaningful biophysical observables often exceeds the limits of conventional imaging. Typical foci have characteristic sizes on the order of hundreds of nanometers, which makes standard fluorescence microscopy constrained by the diffraction limit and insensitive to protein motion within foci. SPT offers a powerful alternative, providing nanometer-scale spatial resolution and millisecond-scale temporal information (11). To fully leverage this capability, however, methods must be developed that allow extraction of fundamental physical observables such as spatial diffusivity, free-energy landscapes, and ultimately fluctuations and potential breakdowns of detailed balance. While diffusion can be reliably inferred using established approaches (19, 20), estimating the underlying energy landscape from SPT data is substantially more challenging. Existing strategies often require densely sampled trajectories (21–26), and are typically optimized for reconstructing global energy profiles rather than quantifying specific physical parameters. Other methods focus on classifying individual tracks as free, directed, or confined diffusion (27–30).

In this paper, we introduce two complementary analytical tools for SPT data aimed at distinguishing focus types. We first present a Hidden Markov Model (HMM) for correcting focus center movement during recording. Building on this, we develop a method that reliably estimates local density and extracts the associated free energy of the underlying focus. We validate both approaches using simulated trajectories and then apply them to five experimental SPT datasets spanning DNA-repair proteins with different expected material states—Rad52 (with one and two DSBs), Rad52-SUMO, Rad51, and Rfa1. Placing each focus in a single diffusivity–density plane, we use the recovered physical fingerprints to classify it as liquid-like (LLPS) or polymer-bridged (PPPS), and compare the outcome against theoretical predictions (13) and the independently expected behaviour of these proteins (11, 14, 31).

MATERIALS AND METHODS

Experimental setup

The SPT data analyzed in this study comprise five DNA-repair conditions: Rad52 with one DSB, Rad52 with two DSBs, Rad52-SUMO, Rad51, and Rfa1, providing a range of DNA-repair-related protein conditions and DSB geometries. The data were either obtained from previously published studies or acquired for the present study using the same experimental procedures. All strains are isogenic RAD5⁺ W303 derivatives.

Strain construction, fluorescent tagging, imaging conditions, and trajectory extraction are described in detail in Mine-Hattab et al. 2021 (14) and the associated references. We summarise only the points relevant to the present analysis. Each repair protein was tagged at its N-terminus with either mMaple or HaloTag using a CRISPR-Cas9 fusion strategy with a short flexible linker (15 or 48 nucleotides). For the HaloTag strains, the pleiotropic drug-resistance gene PDR5 was replaced with selection markers (NAT or LEU2) to allow intracellular delivery of the Halo ligand. The functionality of the tagged proteins was verified by dilution assays on plates containing methyl methanesulfonate (MMS) at varying concentrations, as in the original publication. Rad51-mEOS3.2 was tagged using the strategy described in Liu et al. 2023 (31) and the mEOS3.2 sequence from the plasmid Addgene plasmid # 87030.

Live-cell SPT was performed at 50 Hz ($\Delta t = 20$ ms) with a localization precision of $\sigma_{\text{loc}} \approx 25$ nm. Each cell contributes a set of two-dimensional trajectories, with one focus per nucleus visible in the focal plane (Fig. 1A). The focal depth, which depends on the microscope, wavelength, exposure and detection threshold, is approximately (300 ± 50) nm for the analyzed data sets. The raw trajectory files are publicly available together with the original publication and were used here without modification.

Analysis workflow

The analysis consists of four steps, summarized in Fig. 1C. First, the distribution of frame-to-frame displacements is decomposed into slow and fast diffusive populations associated predominantly with the focus and surrounding nucleoplasm, respectively. Second, motion of the focus during acquisition is inferred using a hidden Markov model (HMM), allowing the SPT localizations to be transformed into a focus-centered reference frame. Third, the corrected localizations are used to estimate the spatial density inside and outside the focus and thereby determine the corresponding partition free-energy difference. Finally, the slow diffusion coefficient is corrected for localization noise using a covariance-based estimator. Together, these quantities provide the density and diffusivity ratios used to characterize each focus.

FLOWCHART

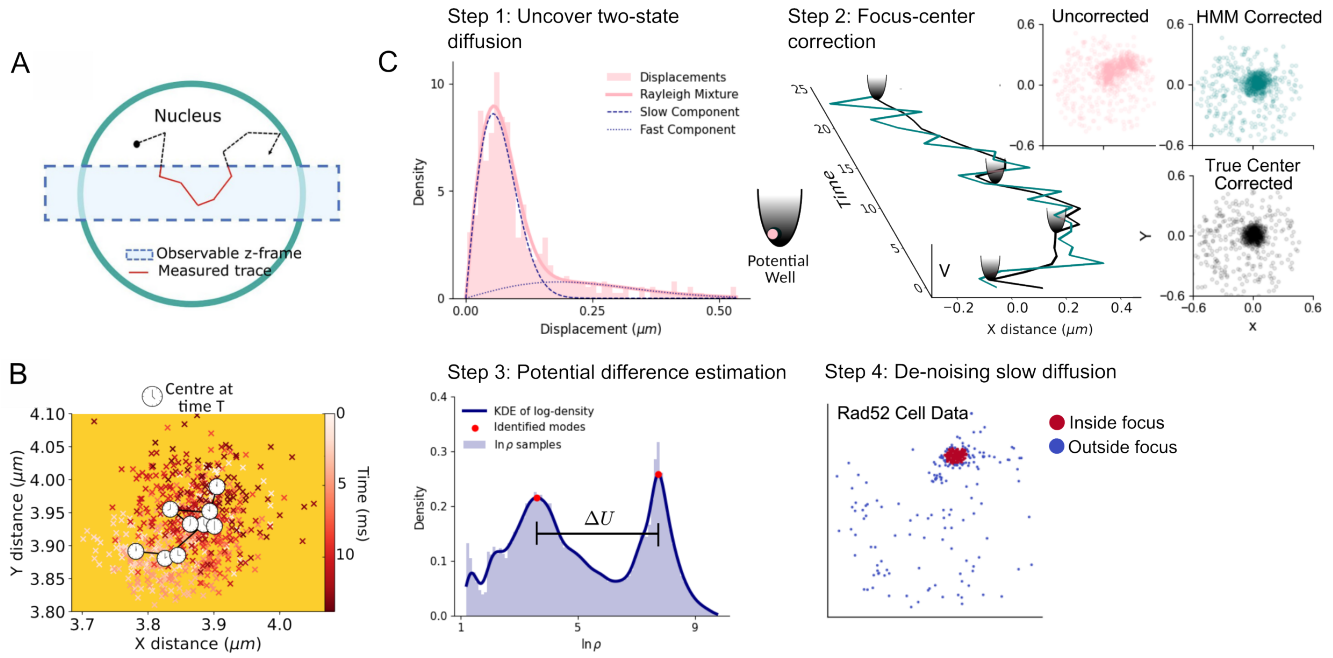


Figure 1: Experimental geometry and workflow for extracting physical observables from sparse SPT data. **A** Schematic of the single-particle tracking (SPT) experiment, in which individual molecules are tracked in two dimensions. The observable (z)-section is indicated by the blue rectangle. **B** Spatial distribution of SPT localizations within a focus, colored according to recording time. Clock symbols indicate the mean position over 2-s intervals, illustrating motion of the average focus center during acquisition. **C** Flowchart of the proposed framework for extracting physical observables from sparse SPT data.

Two-state diffusion estimation

To distinguish molecular mobility inside and outside the focus, we analyze the frame-to-frame displacement lengths, $\Delta r = |\mathbf{r}_{i+1} - \mathbf{r}_i|$. For two-dimensional Brownian diffusion, displacement lengths follow a Rayleigh distribution. We therefore describe the observed displacements using a two-component Rayleigh mixture,

$$f(\Delta r; D_f, D_n, p) = p \frac{\Delta r}{2D_f \Delta t} \exp\left(-\frac{\Delta r^2}{4D_f \Delta t}\right) + (1-p) \frac{\Delta r}{2D_n \Delta t} \exp\left(-\frac{\Delta r^2}{4D_n \Delta t}\right), \quad (1)$$

where D_f and D_n are the diffusion coefficients of the slow and fast populations, respectively, and p is the fraction of observed displacements assigned to the focus-associated population. The two components are interpreted predominantly as focus-associated and nucleoplasmic motion, respectively. This model assumes locally homogeneous diffusion within each compartment. Additional diffusive subpopulations can be incorporated where supported by the experimental data (28, 29).

Focus-center correction

During an SPT recording, the focus itself moves within the imaging plane (Fig. 1B). Pooling all SPT localizations would therefore result in a broadening of the apparent spatial distribution of particles associated with the focus. This motion-induced broadening lowers the apparent focus density and biases the inferred partition free energy. Experiments report focus center motion with diffusion coefficient $D_c \approx 0.005 \mu\text{m}^2/\text{s}$ (14). We therefore infer the time-dependent focus-center position and transform the measured coordinates into a focus-centered reference frame (Fig. 1C, Step 2).

Because the focus-center trajectory is not directly observed, we infer it from the measured particle positions using an HMM adapted from Slator et al. (32). The sigmoidal potential used in the simulations does not yield a tractable transition model, so particle motion within the focus is approximated locally by an overdamped Ornstein-Uhlenbeck (OU) process,

$$\begin{aligned} d\mathbf{X} &= \kappa(\mathbf{C} - \mathbf{X})dt + \sqrt{2D_f} d\mathbf{W}_x(t), \\ d\mathbf{C} &= \sqrt{2D_c} d\mathbf{W}_c(t), \end{aligned} \quad (2)$$

where $\mathbf{X}$ denotes the observed particle position, $\mathbf{C}$ the latent focus-center position, and κ the strength of the harmonic restoring force. The harmonic approximation is used only for center inference and is not intended to describe the focus boundary.

Kalman-RTS inference

Conditional on the OU parameters (κ, D_f, D_c) , Eq. (2) defines a linear-Gaussian state-space model. The latent center, $\mathbf{C}_i$, follows a random walk with process variance $2D_c\Delta t_i$, and each consecutive observed displacement $\mathbf{Y}_i \equiv \mathbf{X}_{i+1} - a_i \mathbf{X}_i$ with $a_i = e^{-\kappa\Delta t_i}$ is a noisy linear measurement of $\mathbf{C}_i$,

$$\mathbf{Y}_i = (1 - a_i) \mathbf{C}_i + \boldsymbol{\varepsilon}_i, \quad \boldsymbol{\varepsilon}_i \sim \mathcal{N}\left(0, \frac{D_f}{\kappa} (1 - a_i^2) \mathbf{I}\right). \quad (3)$$

The posterior over the center trajectory, $\mathbf{C}$, given the full observed trajectory, $\mathbf{X}$, is therefore multivariate Gaussian and can be computed exactly in $O(N)$ operations by a Kalman filter (forward pass) followed by a Rauch–Tung–Striebel smoother (backward pass). We use the posterior mean

$$\hat{\mathbf{C}}_i = \mathbb{E}[\mathbf{C}_i | \mathbf{X}, \kappa, D_f, D_c] \quad (4)$$

as the inferred focus-center trajectory. The full filtering and smoothing recursions are provided in Supplemental Note S1.

Parameter estimation and anchor traces

The OU parameters (κ, D_f) are estimated for each anchor trajectory by maximizing the marginal likelihood, with D_c fixed to the experimentally measured value (Table S1). A Gaussian prior centered on the slow-population diffusion coefficient obtained from the Rayleigh-mixture analysis is used to stabilize the estimate of D_f for short trajectories.

Center inference is restricted to trajectories with $N \geq N_{\min}$ hereafter termed anchor trajectories, because reliable inference requires the particle to remain associated with the focus over several frames. **The estimated partition free energy is insensitive to $N_{\min}$ over the tested range $5 \leq N_{\min} \leq 20$ (Fig. S1); we therefore use $N_{\min} = 10$. Each anchor trace produces a per-frame center estimate $\hat{\mathbf{C}}_i$. Localizations from shorter trajectories are corrected using the temporally nearest available anchor-center estimate. The complete inference procedure is described in Supplemental Note S1.**

Local-density and partition free-energy estimation

At equilibrium, the densities inside and outside the focus are related to the corresponding free-energy difference through the Boltzmann relation (33, 34),

$$\exp\left(\frac{\epsilon_f - \epsilon_n}{k_B T}\right) = \frac{\rho_n}{\rho_f}, \quad (5)$$

where ρ_f and ρ_n denote particle densities in the focus and nucleoplasm, respectively, and the free energy difference is $\Delta U = \epsilon_n - \epsilon_f$. Thus, estimating the density ratio directly provides the partition free-energy difference without requiring an explicit estimate of the focus or nuclear volume. Previous approaches inferred these densities from the Rayleigh mixture fraction p and compartment volumes (13), making the result sensitive to estimates of focus and nuclear size.

We estimate the densities from the pooled, focus-center-corrected localizations using the local-density sampling procedure illustrated in Fig. 2A and Fig. 1C, Step 3. Sampling disks with variable radii are distributed throughout the recorded region. The area of each disk should be small enough to sample only local regions that are accessible to the particles, but large enough to include a sufficient number of particle recordings. Within each sampling disk i , we count the number of particles, X_i , and estimate the local density as the particle count divided by the disk area, A_i : $\hat{\rho}_i = X_i/A_i$. If the particles are locally distributed with density (ρ) , the particle count within a disk is assumed to follow a Poisson distribution with mean $(\lambda_i = \rho A_i)$. Thus, the expected count, $\mathbb{E}[X_i]$, scales with the area of the sampling disk, while normalization by A_i places all local density estimates on the same density scale:

$$\rho = \frac{\mathbb{E}[X_i]}{A_i}. \quad (6)$$

Varying the disk radius prevents the discrete Poisson counts from producing a discrete set of density estimates: although a fixed radius yields values X_i/A_i , variable radii generate a quasi-continuous distribution of $\hat{\rho}_i$. Zero-count sampling disks are resampled by re-drawing disk radii until a SPT localization is detected or a maximum number of re-draws have occurred.

The resulting distribution of local ln-densities, $\ln(X_i/A_i)$, is typically bimodal, with modes associated with the focus and surrounding nucleoplasm. We estimate the partition free energy from the separation between these modes,

$$\frac{\Delta U}{k_B T} = \ln \rho_{\text{in}} - \ln \rho_{\text{out}} \approx \text{mode}(\ln \hat{\rho})_{\text{in}} - \text{mode}(\ln \hat{\rho})_{\text{out}}. \quad (7)$$

as illustrated in Fig. 1C, Step 3. The use of the modal rather than mean density is derived in Supplemental Note S3. A brief motivation for using the mode of the distribution is the following. In a region where the underlying density is constant at ρ , each disk count is $X_i \sim \text{Poisson}(\rho A_i)$. This means the per-disk estimator is $\hat{\rho}_i = X_i/A_i$ with expectation ρ and standard deviation $\sqrt{\rho/A_i}$.

Localization-noise correction of the slow diffusion coefficient

Localization uncertainty has a substantial relative effect on the slow diffusion coefficient inferred from the Rayleigh mixture, while leaving the fast diffusion coefficient relatively unaltered. At $\Delta t = 20\text{ms}$, a molecule with $D_f \approx 5 \times 10^{-2} \mu\text{m}^2/\text{s}$ has a characteristic displacement of approximately 40 nm, comparable to the experimental localization precision of approximately 25 nm. We therefore correct the slow-component diffusion coefficient using the covariance-based estimator of Vestergaard et al. (19). This method separates the two contributions (diffusion and experimental noise) by using the fact that the localisation error is white while the displacement of a Brownian walker is correlated only through the shared endpoint of consecutive steps. For two-dimensional Brownian motion with independent localization error of variance σ^2 , consecutive measured displacements satisfy

$$\text{Var}(\Delta x_n) = 2D_f \Delta t + 2\sigma^2, \quad \text{Cov}(\Delta x_n, \Delta x_{n+1}) = -\sigma^2, \quad (8)$$

because the localization error at their shared endpoint enters the two consecutive displacements with opposite signs. Combining the corresponding x - and y -components gives

$$2\sigma^2 = -\langle \Delta x_n \Delta x_{n+1} + \Delta y_n \Delta y_{n+1} \rangle, \quad b_1^2 = \frac{1}{2} \langle \Delta x_n^2 + \Delta y_n^2 \rangle, \quad (9)$$

from which the localization-noise-corrected slow diffusion coefficient is obtained as

$$D_f = \frac{b_1^2 - 2\sigma^2}{2\Delta t}. \quad (10)$$

Here b_1 is the estimate of the Rayleigh distribution of the single displacement distance.

Simulation model for validation

To validate the analysis pipeline against a known ground truth, we simulate particle motion in a diffusing focus using the overdamped Langevin equations

$$d\mathbf{r} = \left[\nabla D(\mathbf{r}, \mathbf{r}_c) - \frac{D(\mathbf{r}, \mathbf{r}_c)}{k_B T} \nabla U(\mathbf{r}, \mathbf{r}_c) \right] dt + \sqrt{2D(\mathbf{r}, \mathbf{r}_c)} d\mathbf{W}_b(\mathbf{t}), \quad (11)$$

$$d\mathbf{r}_c = \sqrt{2D_c} d\mathbf{W}_c(\mathbf{t}).$$

Here $\mathbf{r}$ is the particle position, $\mathbf{r}_c$ is the focus-center position, and $d\mathbf{W}_b(t)$ and $d\mathbf{W}_c(t)$ are independent three-dimensional Wiener processes. The term $\nabla D(\mathbf{r}, \mathbf{r}_c)$ is a noise-induced drift term arising under the Itô convention. We model the spatially varying particle diffusion coefficient, $D(\mathbf{r}, \mathbf{r}_c)$, and the potential, $U(\mathbf{r}, \mathbf{r}_c)$, by sigmoidal functions:

$$D(\mathbf{r}, \mathbf{r}_c) = D_f + \frac{D_n - D_f}{1 + e^{-\beta(|\mathbf{r}-\mathbf{r}_c|-r_f)}}, \quad U(\mathbf{r}, \mathbf{r}_c) = \frac{v}{1 + e^{-\beta(|\mathbf{r}-\mathbf{r}_c|-r_f)}}. \quad (12)$$

Here, D_f and D_n are the diffusion coefficients inside the focus and in the surrounding nucleus, respectively, r_f the focus radius, v the potential difference between the focus interior and the surrounding nucleus, and β controls the steepness of the transition at the focus boundary. Parameter were chosen to reproduce characteristic properties of experimental DSB foci and are listed in Table S1.

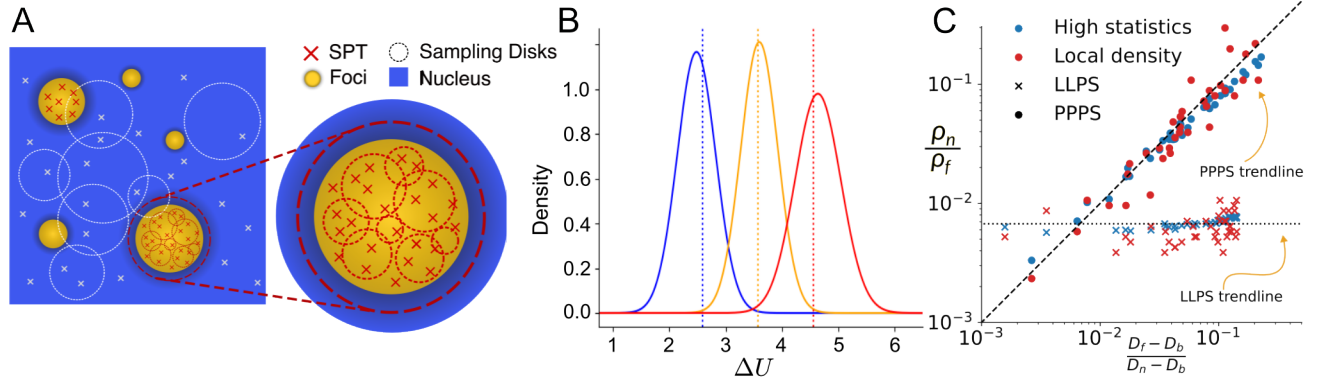


Figure 2: Validation of the local-density sampling method using simulated SPT data. **A** Schematic of the nucleus, focus, and SPT localizations. Dotted circles denote sampling disks of different sizes used to estimate the local density. **B** Distributions of measured ΔU for different underlying values of ΔU_{True} . Dotted lines indicate the simulated true values, and solid curves show Gaussian fits to the sampled data for $N_{\text{repeat}} = 100$ repeats. All distributions have a width of $\sigma \approx 0.35 k_B T$. **C** Comparison of a high-statistics method and the local-density sampler for simulated LLPS and PPPS foci. Markers indicate the underlying focus type used to generate the sampled data, and dotted lines show the corresponding focus-type trends. The particle densities were $\rho \approx 250 \mu\text{m}^{-2}$ for the local-density method and $\rho \approx 11,000 \mu\text{m}^{-2}$ for the high-statistics method. Simulation parameters for the different focus models are listed in Table S1.

RESULTS

Validation of the local density sampler

To validate the local-density sampler, we apply it to simulated data with a known potential difference between the focus and the surrounding nucleus. Three-dimensional SPT trajectories are generated using Eqs. 11 and 12. To mimic the experimental measurements, localizations outside the focal plane are discarded and Gaussian localization noise with standard deviation $\sigma_{\text{loc}} = 25 \text{ nm}$ is added to the retained (x, y) -coordinates. Because the experiment observes a finite-depth projection of a three-dimensional system, the measured density differs from the underlying 3D equilibrium density. Let $s = \sqrt{x^2 + y^2}$ be the recorded 2D radial coordinate. A localization s could originate from any unresolved axial position z in the focal slab. The expected observed 2D localization density is therefore

$$\rho_{2D}(s) = C \int_{-Z_L}^{Z_L} \exp \left[-U \left(\sqrt{s^2 + z^2} \right) \right] dz, \quad (13)$$

where U is the potential and Z_L is the focal half-depth. For $Z_L \approx r_f$ and a steep focus-boundary transition (large β), the difference between the true 3D equilibrium density and the apparent 2D density is small. We evaluate $\rho_{2D}(s)$ with the sigmoidal potential in Eq. 12 and define $\Delta U_{2D}/k_B T = \ln \rho_{2D}(0) - \ln \rho_{2D}(r_n)$ as the target potential difference to be recovered by the sampler. ΔU_{2D} as a function of Z_L is shown in Fig. S3, along with the relative error which can arise with imprecise estimation of the focal depth.

We test three underlying potential differences, $\Delta U/k_B T \in [3.0, 4.0, 5.0]$, corresponding to projected values $\Delta U_{2D}/k_B T \in [2.6, 3.6, 4.6]$ for $Z_L = 0.15 \mu\text{m}$. Using bootstrapping with $N_{\text{repeat}} = 100$ realizations, we recover Gaussian distributions of estimated potential differences centered on the target values, with fitted widths of approximately $\sigma_{\Delta U} \approx 0.35 k_B T$ (Fig. 2B).

Additionally, we test the local-density sampler in combination with diffusivity measurements and test whether the two focus models can be distinguished in the diffusivity–density plane. For the PPPS model, the diffusion coefficient and potential difference are related by $\Delta U/k_B T = \ln[(D_n - D_b)/(D_f - D_b)]$, where $D_b = 0.005 \mu\text{m}^2/\text{s}$ is the diffusion coefficient of the PPPS binding sites (Supplemental Note S2). By contrast, for the LLPS model, the density ratio, ρ_n/ρ_p , remain constant as the diffusivity contrast is varied. At fixed ΔU , we simulate trajectories for the LLPS model using Eqs. 11 and 12 and for the PPPS model using the framework introduced previously (13). The density ratio is estimated using the local-density sampler (Red markers Fig. 2C), whereas the diffusivity ratio is obtained from the two-component Rayleigh-mixture fit in Eq. 1. Note that for all cell types considered in the following, we tested 1, 2 and 3 component Rayleigh-mixture fit (See Supplementary Note S5). We conclude that the two-component Rayleigh-mixture model works to a appreciable level of certainty. The local-density sampler reproduces the expected LLPS and PPPS trends at substantially lower localization densities than the previously proposed high-statistics method (Blue markers Fig. 2C).

Validation of the focus-center correction

We next validate the HMM-based correction for focus motion. Single-particle trajectories are simulated according to Eqs. 11 - 12, while allowing the focus center to diffuse with diffusion coefficient D_c . The focus-center trajectory is then inferred from the simulated SPT data using the HMM. The inferred focus-center trajectory closely follows the true trajectory used in the simulation (Fig. 3A). For each localization, we subtract the inferred center position $\hat{\mathbf{C}}(t)$, thereby transforming the trajectory into the estimated focus-centered reference frame: $\mathbf{x}_c = \mathbf{x} - \hat{\mathbf{C}}$. Since the HMM attempts to resample for a harmonic potential, the consequence is contraction of the corrected radial distribution. Comparing the corrected radial density profile with the true focus-centered profile, we find empirically

$$\rho_c(r) \simeq \rho_{\text{true}}(r) e^{-\alpha r^2}, \quad \alpha \approx 2 \times 10^2 \mu\text{m}^{-2}, \quad (14)$$

corresponding to a spurious soft-harmonic confinement of width $1/\sqrt{2\alpha} \approx 45$ nm on top of the true profile (Fig. 3B). Despite this distortion of the radial profile, its effect on the inferred partition free energy is limited. The local-density estimator determines the potential difference from the separation between the two modes of the ln-density distribution,

$$\frac{\Delta U}{k_B T} = \ln \left(\frac{\rho_{\text{in}}}{\rho_{\text{out}}} \right) = \ln \rho(0) - \ln \rho_{\text{out}}, \quad (15)$$

where $\rho(0)$ is the modal density of the focus-associated population and ρ_{out} is the nucleoplasmic background density. At the center, the additional HMM-induced factor satisfies $e^{-\alpha r^2}|_{r=0} = 1$, leaving the central density unchanged. The background distribution may undergo a small shift during sampling but is not reshaped, so its mode is likewise preserved. Consequently, the separation between the two density modes remains largely insensitive to the HMM-induced change in the radial profile (Fig. 3B; see Supplemental Note S3 for the derivation of the maximal density).

Finally, we test the complete correction-and-sampling procedure as a function of data availability. After focus-center correction, the local-density sampler recovers the known potential difference within the estimated uncertainty, with precision improving as the mean number of detections increases. Compared with the confined-MSD method used previously (13) and a KDE-based estimate, the local-density sampler remains closer to the true potential and shows lower variability across the tested detection densities (Fig. 3D).

Together, these simulations show that the combined framework of HMM-based focus-center correction and local-density subsampling enables reliable extraction of the potential difference between a moving focus and its surroundings, even when the available SPT data are sparse. We next apply this combined framework to experimental SPT data.

Application on SPT data

We next demonstrate how the combined focus-center correction and local-density sampling framework can be used to extract quantitative observables from experimental data (visually described in the flowchart of Fig. 1C). We analyzed SPT trajectories for five protein conditions: Rad52 at a single DSB, Rad52 at two DSBs, Rad52-SUMO, Rad51, and Rfa1.

We restricted the analysis to nuclei containing sufficient SPT localizations to be treated individually. For each cell, the HMM was used to infer and subtract the motion of the focus center. The local-density sampler was then applied to the focus-centered trajectories to reconstruct the distribution of local ln densities inside and outside the focus (Fig. 4A). Although neither compartment was spatially uniform, the sampled local-density distribution was clearly bimodal for most cells (Fig. 1C, third step of the flowchart). We therefore estimated the potential difference, ΔU , from the separation between the two modes, as defined in Eq. 7. In parallel, a two-component Rayleigh-mixture model (Eq. 1) was fitted to the displacement distribution from the same cell to estimate the focus and nucleoplasmic diffusion coefficients, D_f and D_n , respectively. The slow component was corrected for localization noise as described in the methods section. Since the theoretical treatment of the potential assumes an equilibrium distribution, we wanted to validate this assumption. Fundamentally, a system that undergoes DNA repair is by definition out of equilibrium, but the investigated relation between diffusivity and potential is applied to the local, focus-conditional distribution of a tracked protein, not to the nucleus as a whole. It therefore requires only that a molecule samples the focus interior while this is stationary to a good approximation. This means that our data need to be in local equilibrium over the short window we observe, not a global nuclear equilibrium. The relevant timescale is the intra-focus relaxation time: $\tau_{\text{relax}} \sim \frac{r_f^2}{D_f} \approx 0.2$ s i.e. only ~ 10 frames at $\Delta t = 20$ ms. Taken in relation to the minute-hour timescales of the repair processes itself, we would assume that we within the short window would see molecules in a quasi-equilibrium. To further validate this, we searched for non-equilibrium fluxes, inspired by Battle et al. (35) (See Methods). We first investigated a gradient flux, which would indicate a non-stationary system as $\nabla \cdot \mathbf{j} \neq 0$. Based on our numerical simulation in equilibrium, we

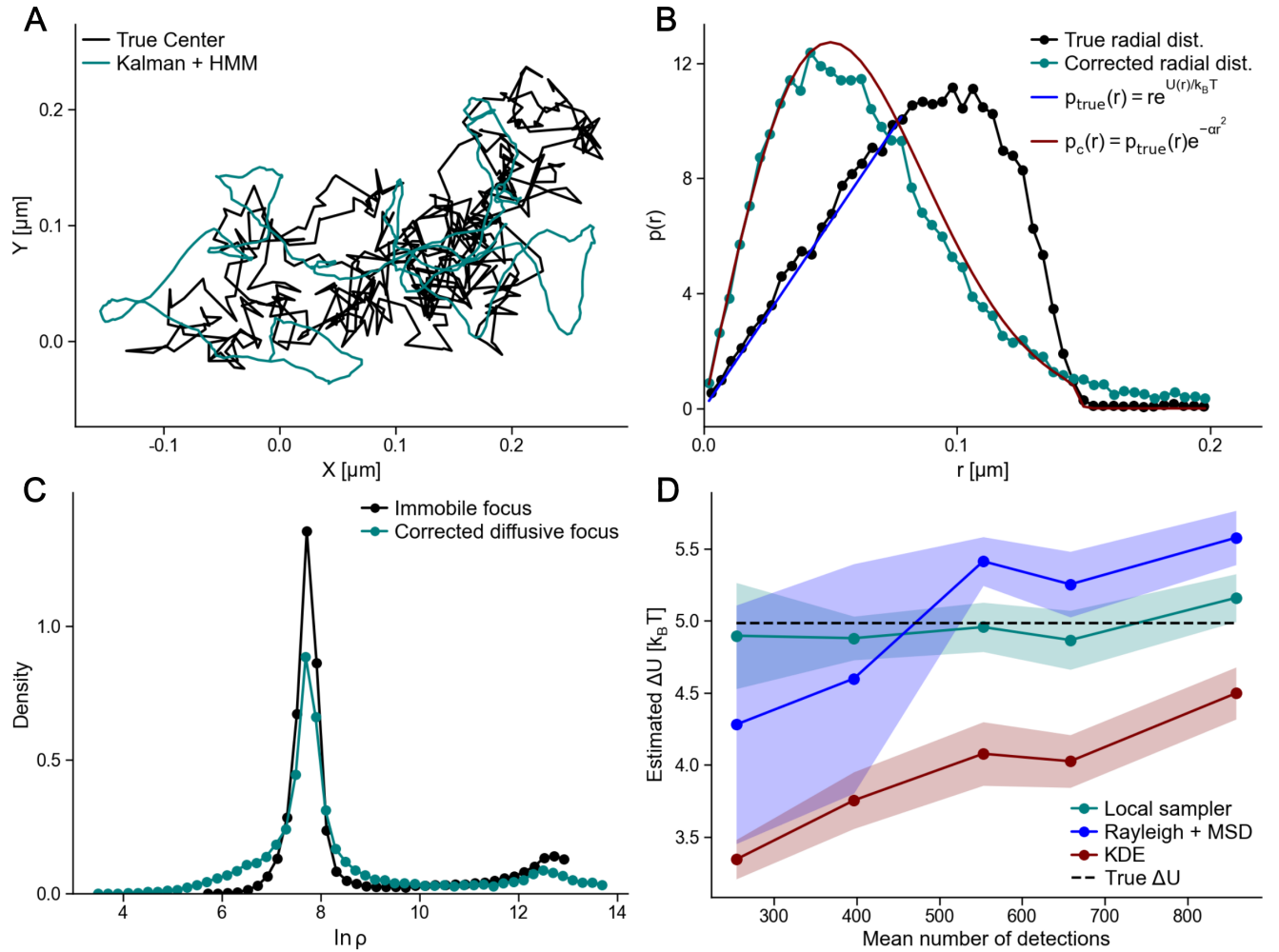


Figure 3: Validation of the HMM-based focus-center correction **A** Comparison of the true and the inferred focus-center trajectory for single-particle simulations using Eqs. 11-12. The black line indicates the true center trajectory used in the simulation. **B** The true radial distribution and the observed radial distribution after the focus-center correction. The HMM-correction reshapes the true radial distribution function by a factor of $e^{-\alpha r^2}$. **C** Measured $\ln$ -density distributions for simulations with an immobile focus and with a diffusive focus whose motion has been corrected for by the HMM-framework. **D** Comparison of three different methods for estimating the potential difference as a function of the mean number of SPT detection using simulated data with a known potential difference. Error bars are standard error.

measured the average distance between the first and the last positions of the in each trace to find the expected variance and 95% confidence level on the data. Next we applied this measure to all cells for all cell types, and compared the observed median to the expected confidence level of the simulation. Performing this analysis showed that no cells could reject our null-hypothesis that the system is in equilibrium (Fig. 4B). Finally, we measured the rotational current, by taking the curl of each trace and by again finding the expected confidence level in the simulation and comparing this to data, we could also conclude that the null-hypothesis of equilibrium could not be rejected (Fig. 4C). Based on this, we can therefore assume that our fundamental assumptions hold.

Each cell was subsequently represented in the diffusivity–density plane introduced in Fig. 2C and shown on logarithmic axes in Fig. 4D. The abscissa is the reduced diffusivity ratio $x = (D_f - D_b)/(D_n - D_b)$, and the ordinate is the measured density ratio $y = \rho_{\text{out}}/\rho_{\text{in}} = \exp(-\Delta U/k_B T)$. The PPPS and LLPS models make distinct predictions in this plane. For a PPPS focus, enrichment is determined entirely by the change in mobility across the focus boundary, $\Delta U = k_B T \ln[(D_n - D_b)/(D_f - D_b)]$ (Note S2). It follows that $y = x$, such that an ideal PPPS focus lies on the identity line in Fig. 4D. By contrast, an LLPS focus contains an additional non-diffusive binding contribution. This produces a potential depth larger than the PPPS prediction and places the corresponding point below the identity line. To quantify this displacement, we defined for each cell the signed logarithmic gap

$$g_i \equiv \ln x_i - \ln y_i = \frac{\Delta U_i - \Delta U_{\text{PPPS},i}}{k_B T} \quad (16)$$

Thus, $g_i = 0$ corresponds to the PPPS prediction, whereas $g_i > 0$ indicates an excess partitioning free energy that is not explained by the diffusivity contrast. Because x_i and y_i are calculated for the same cell before evaluating g_i , this paired measure retains the variability of both observables and their covariance. For each protein condition, we tested the null hypothesis $H_0 : \langle g \rangle = 0$ against the one-sided alternative $H_1 : \langle g \rangle > 0$ using a one-sample t -test on the cell-level values of g_i . The resulting p -values were adjusted across the five conditions using the Holm–Bonferroni procedure. A condition was considered significantly displaced below the PPPS line when the Holm-adjusted p -value was below 0.05. Conditions that did not meet this criterion were considered statistically consistent with the PPPS prediction.

Condition	n	$x \pm \text{SE}_x$	$y \pm \text{SE}_y$	$g [k_B T]$ (95% CI)	p_{Holm}	% corrected localizations
Rad52 (1 DSB)	26	0.076 ± 0.008	0.025 ± 0.012	$+2.76 [+1.96, +3.55]$	$< 10^{-3}$	91.5%
Rad52 (2 DSB)	10	0.163 ± 0.067	0.029 ± 0.018	$+3.00 [+1.54, +4.46]$	0.002	90.6%
Rad52-SUMO	9	0.062 ± 0.013	0.016 ± 0.011	$+2.33 [+1.28, +3.38]$	0.002	64.7%
Rad51	35	0.134 ± 0.027	0.135 ± 0.019	$-0.26 [-0.90, +0.37]$	0.800	47.9%
Rfa1	20	0.051 ± 0.024	0.035 ± 0.013	$+0.52 [-0.44, +1.48]$	0.273	91.0%

Table 1: Per-condition positions in the diffusivity–density plane and statistical test of displacement from the PPPS identity line. The x and y columns report the mean $\pm$ SE of the cell-level reduced diffusivity and density ratios, respectively. The excess energy measures the signed logarithmic displacement below the identity line. Confidence intervals are 95% intervals for the mean g , and p_{Holm} denotes the Holm-corrected one-sided p -value for $H_0 : \langle g \rangle = 0$ against $H_1 : \langle g \rangle > 0$. The last column provides the mean percentage of corrected localizations, which includes both localizations in the anchor traces and localizations corrected by nearest-anchor assignment.

The five conditions separated into two groups according to this criterion (Fig. 4E and Table 1). The large markers and error bars in Fig. 4D show the per-condition mean $\pm$ SE. All three Rad52 conditions were significantly displaced below the identity line after correction for multiple comparisons. Rad52 at one DSB had a mean excess energy of $2.76k_B T$ (95% CI: 1.96 – $3.55k_B T$), Rad52 at two DSBs had a mean excess energy of $3.00k_B T$ (95% CI: 1.54 – $4.46k_B T$), and Rad52-SUMO had a mean excess energy of $2.33k_B T$ (95% CI: 1.28 – $3.38k_B T$). These results indicate that the density enrichment of the Rad52 conditions cannot be explained by their diffusivity contrasts alone and support an additional LLPS-like contribution.

By contrast, neither Rad51 nor Rfa1 differed significantly from the PPPS identity line. Rad51 had a mean excess energy of $-0.26k_B T$ with a confidence interval spanning zero, whereas Rfa1 had a mean excess energy of $0.52k_B T$, also with a confidence interval spanning zero. Their measured partitioning was therefore statistically consistent with enrichment generated by the diffusivity contrast alone.

This separation agrees with previous characterizations of the proteins. Rad51 has been proposed to organize repair foci through polymer bridging (31), while Rfa1/RPA has previously been used as a PPPS reference (11); both observations are consistent with their positions near the PPPS identity line. Conversely, Rad52 foci have been characterized as liquid-like (13, 14), consistent with the significant excess partitioning energies observed for all three Rad52 conditions.

Taken together, the framework identifies a significant LLPS-like contribution to the enrichment of the three Rad52 conditions, while the Rad51 and Rfa1 measurements remain statistically consistent with the PPPS prediction. These classifications are obtained directly from the SPT trajectories without requiring estimates of the focus or nuclear volumes and agree with independent characterizations of the corresponding protein assemblies (11, 13, 14, 31).

DISCUSSION

In this work, we present a new framework for enhancing the measurement of observables in SPT data. Specifically, we develop a method capable of quantifying the partition free energy between a focus and the surrounding nucleoplasm from sparse SPT data. This approach enables the analysis of individual foci, allowing us to investigate heterogeneity within a single cell. Previously, free-energy differences have been quantified by comparing Boltzmann factors, which requires precise estimates of spatial profile of the condensate, typically extracted from the saturation curves of a confined MSD. By employing a local subsampling strategy, we obtain accurate density measurements while making minimal assumptions about the underlying data. This allows us to analyze data sets containing substantially fewer localizations than typically required, making it possible to study individual foci in detail.

Our method relies on local sampling rather than the more common use of global estimators. This improves both the robustness and precision of the measurements while keeping model assumptions minimal. Although related approaches exist for estimating energy landscapes from SPT data, our method is designed for high-precision quantification and is well suited for detecting subtle features that advance our understanding of intracellular organization. These improvements also make it feasible to evaluate cells individually, thereby providing insights into heterogeneity across single-cell systems.

While this study reproduces previously reported results, the strength of the method lies in its potential application to more challenging systems. When the free-energy difference increases, trajectories tend to be longer, and the resulting motion of the focus introduces measurable biases. At the same time, the density outside the focus decreases, making accurate normalization difficult because only a fraction of the nuclear volume is experimentally accessible. Our approach circumvents these limitations and is applicable to a broader class of data sets. A natural extension of this work would be to incorporate a similar local-subsampling strategy to generate spatially resolved diffusivity maps, offering a pathway to uncover additional hidden structure within the nucleus.

CONCLUSION

In summary, this paper introduces a method for determining spatial density from sparse SPT data and subsequently converting these densities into free-energy potentials. This represents a significant improvement over existing techniques and opens the door to extracting and interpreting increasingly detailed spatio-temporal information from intracellular SPT experiments.

DATA AVAILABILITY

A python implementation of the tools can be found at the repository: <https://doi.org/10.5281/zenodo.22121679>

In the paper, we have used a previously published dataset of Rad52 SPT data (14). Source data files are available on zenodo using the following link: <https://zenodo.org/record/4495116>.

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AUTHOR CONTRIBUTIONS

M.S.N, E.J and M.S.H designed the research. M.S.N, E.J and M.S.H carried out simulations and analyzed the data. E.C and J.M.H performed the experiments. M.S.N, E.J, E.C M.H.J, J.M.H and M.S.H wrote the article.

DECLARATION OF INTERESTS

The authors declare no competing interests.

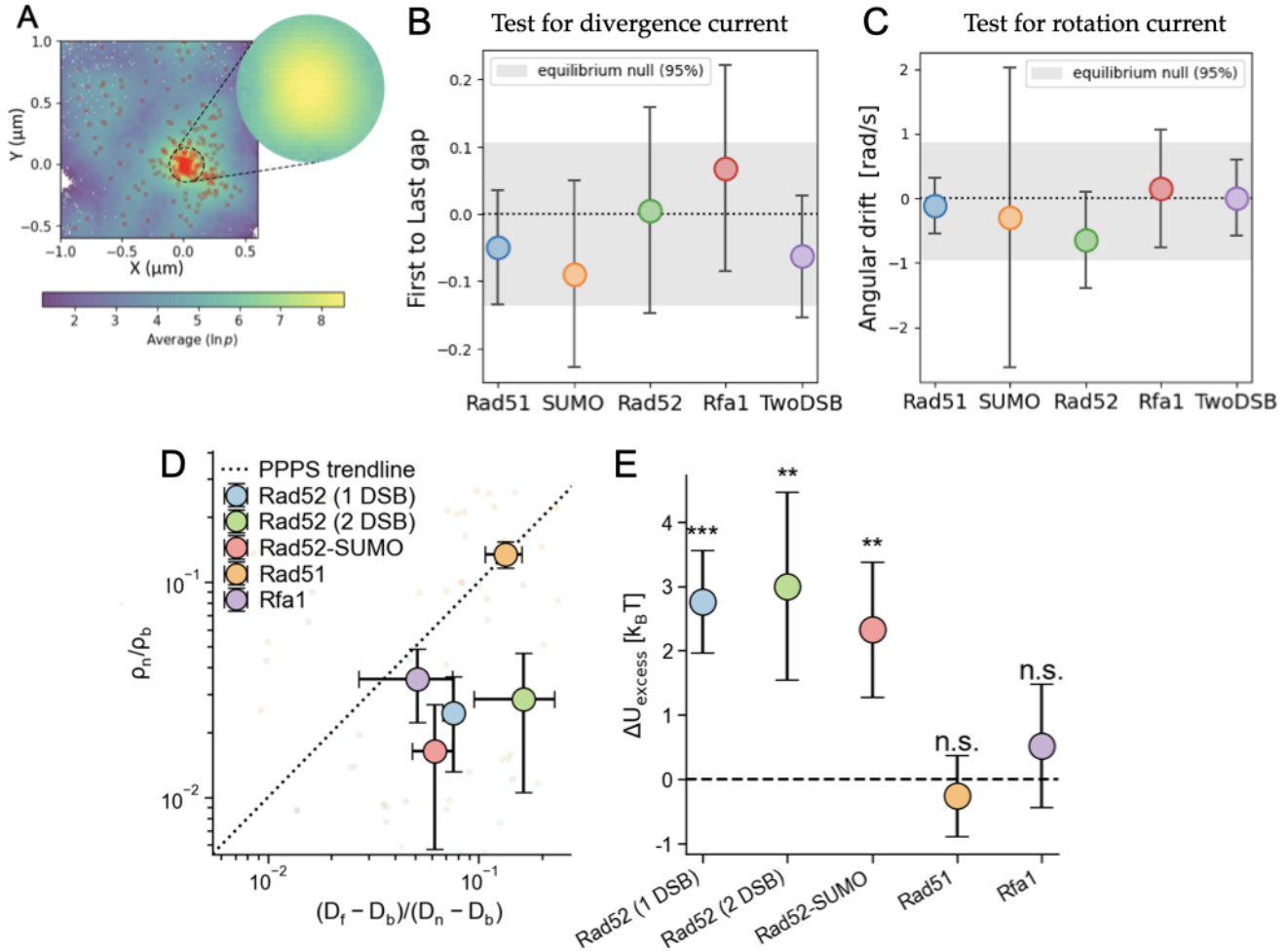


Figure 4: Application of the combined focus-center correction and local-density sampling framework on five difference experimental SPT datasets. **A** Mean ln-density map constructed by the local-density sampler for a representative experimental Rad52 (one DSB) recording. Mean ln-densities were calculated using sampling disks centered on a grid. Blank regions indicate grid points for which no density could be estimated because no data were recorded. **B** Measured difference between the average first position in the trace and the last position in the trace, done for all cells. Grey line represent the simulated 95% confidence level expected for an equilibrium system. **C** Measured rotational current measured as the mean and curl of each individual trace. Grey line represent the simulated 95% confidence level expected for an equilibrium system. **D** Density ratio $\rho_n/\rho_f = \exp(-\Delta U/k_B T)$ versus the reduced diffusivity ratio $(D_f - D_b)/(D_n - D_b)$ (logarithmic axes). The dotted line is the PPPS prediction $\rho_n/\rho_f = (D_f - D_b)/(D_n - D_b)$. Foci below it carry excess binding energy beyond the diffusivity contrast, which is a signature of LLPS. Faint dots are individual cells, while large markers are the per-protein mean and error bars denote standard error. **E** Classification of the five proteins as either LLPS or PPPS by quantifying the excess binding energy: $\Delta U_{\text{excess}} \equiv \ln(D_f - D_b)/(D_n - D_b) - \ln \rho_n/\rho_f$. ΔU_{excess} is calculated per-cell and error bars denote 95% CI.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

During the preparation of this work the authors used ChatGpt and Claude to assist with editing text and with the verification and optimization of analysis code. After using this tool, the authors reviewed, tested, and edited all content and code, and take full responsibility for the content of the publication.

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SUPPLEMENTAL INFORMATION

An online supplement to this article can be found by visiting BJ Online at <http://www.biophysj.org>.

Note S1: Center correction for motion of the focus

To correct for motion of the focus during image acquisition, we infer the time-dependent position of the focus center and transform particle coordinates into a focus-centered reference frame. Our model is an adaptation of a confinement-based hidden-state framework by Slator et al. 2018 (32).

The original framework allows a tracked particle to switch between freely diffusing and confined states. Here, we only use trajectories that are already classified as long-lived and slowly diffusing within the focus. Consequently, we do not infer transitions between free and confined states. Instead, the only hidden variable is the two-dimensional position of the moving focus center. As shown below, restricting the hidden state in this way makes the model linear and Gaussian throughout, so that the focus-center path can be inferred exactly by Kalman filtering and smoothing, without the Markov chain Monte Carlo sampling required in the general framework.

The model is applied to the projected coordinates recorded in the focal plane. Thus, it describes the in-plane motion of the focus and assumes that diffusion in the z -direction is negligible. Tracks that leave the focal plane or escape the focus are excluded by the trajectory-selection criteria described in the main text.

Consider a particle trajectory consisting of $N + 1$ recorded particle positions,

$$\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_N, \quad (17)$$

at times $t_0, t_1, \dots, t_N$, where $\mathbf{x}_i \in \mathbb{R}^2$. The corresponding, unobserved focus-center positions are denoted by $\mathbf{c}_0, \mathbf{c}_1, \dots, \mathbf{c}_N$, with $\mathbf{c}_i \in \mathbb{R}^2$. The time interval between two consecutive observations is $\Delta t_i = t_{i+1} - t_i$. We model a particle confined within the focus by an overdamped Ornstein-Uhlenbeck (OU) process which is defined by the stochastic differential equation,

$$d\mathbf{X}_t = \kappa (\mathbf{C}_t - \mathbf{X}_t) dt + \sqrt{2D_f} d\mathbf{W}_t^X \quad (18)$$

where D_f is the particle diffusion coefficient within the focus, κ is the confinement or relaxation rate, and $\mathbf{W}_t^X$ is a two-dimensional Wiener process. Equation 18 is equivalent to motion in a harmonic potential. This harmonic potential provides a tractable local approximation that captures the two features needed for center inference: particles fluctuate around the focus center, while the center itself can move over time. The focus center is modeled as a Brownian random walk,

$$d\mathbf{C}_t = \sqrt{2D_c} d\mathbf{W}_t^C, \quad (19)$$

where D_c is the focus center diffusion coefficient and $\mathbf{W}_t^C$ is an independent two-dimensional Wiener process. In the present analysis, D_c is fixed to the experimentally measured value rather than inferred separately from each particle trajectory.

For each acquisition interval, we use a frozen-center approximation: the focus center is treated as constant while the particle evolves from t_i to t_{i+1} , and the center displacement is then represented by a random-walk transition. This approximation is appropriate because the root-mean-square displacement of the center during one frame, $\sqrt{2D_c \Delta t}$, is substantially smaller than the characteristic focus radius for the parameter range considered here. With the center held at $\mathbf{c}_i$ during the interval $[t_i, t_{i+1}]$, Eq. 18 has the exact OU transition density

$$\mathbf{X}_{i+1} \Big|_{\mathbf{X}_i} \approx \mathcal{N} \left(\mathbf{C}_i + (\mathbf{X}_i - \mathbf{C}_i) e^{-\kappa \Delta t_i}, \frac{D_f}{\kappa} (1 - e^{-2\kappa \Delta t_i}) \right) \quad (20)$$

where $\mathcal{N}(x; \mu, \sigma)$ is a normal distribution with mean μ and variance σ . The focus center transition density is given by

$$\mathbf{C}_{i+1} \Big|_{\mathbf{C}_i} \approx \mathcal{N}(\mathbf{C}_i, 2D_c \Delta t_i), \quad (21)$$

as we assume it moves like a random walker. Thus, the conditional mean of the particle position is a weighted average of the previous particle position and the current center position.

Equations 20 and 21 define a continuous-state hidden Markov model. The particle positions $\mathbf{x}_i$ are observed, whereas the center positions $\mathbf{c}_i$ are hidden. Conditional on the current center position, the next particle position is independent of earlier particle positions. Similarly, the center follows a Markov random walk.

Let $\theta = (\kappa, D_f, D_c)$ denote the model parameters. Because D_c is fixed from independent measurements, the quantities inferred for each accepted trajectory are κ , D_f , and the latent center path $\mathbf{c}_{0:N}$. The posterior distribution is defined with Bayes'

rule as the product of the likelihood of observing $\mathbf{x}_i$ given the parameters θ and the prior $\pi(\theta, \mathbf{C}_0)$ describing our initial guess for the system:

$$\pi(\theta, \mathbf{C}|\mathbf{X}) \propto \pi(\theta, \mathbf{C}_0) \prod_{i=1}^N \mathcal{N}\left(\Delta \mathbf{X}_i; \mathbf{C}_i - \mathbf{X}_i + (\mathbf{X}_i - \mathbf{C}_i)e^{-\kappa \Delta t_i}, \frac{D_f}{\kappa}(1 - e^{-2\kappa \Delta t_i})\right) \prod_{i=1}^{N-1} \mathcal{N}(\Delta \mathbf{C}_i; 0, 2D_c \Delta t_i) \quad (22)$$

The first product is the likelihood of the observed particle trajectory conditional on a proposed center path. The second product is the prior probability of the center path under Brownian motion.

Exact inference by Kalman filtering and smoothing

Because the focus center is the only hidden variable and every density in Eq. 22 is Gaussian, the model is a linear Gaussian state-space model. The posterior over the center path is therefore itself Gaussian, and the mean, which is the quantity we need for the coordinate transformation, can be computed in closed form by Kalman filtering and smoothing, with no Monte Carlo sampling and hence no run-to-run sampling variability.

To make the dependence on the hidden state explicit, we rewrite the particle transition (Eq. 20) as a linear observation of the center. Defining the one-step quantity

$$\mathbf{Y}_i = \mathbf{X}_{i+1} - e^{-\kappa \Delta t_i} \mathbf{X}_i, \quad (23)$$

Eq. 20 becomes

$$\mathbf{Y}_i = H_i \mathbf{C}_i + \boldsymbol{\varepsilon}_i, \quad H_i = 1 - e^{-\kappa \Delta t_i}, \quad \boldsymbol{\varepsilon}_i \sim \mathcal{N}(0, R_i), \quad R_i = \frac{D_f}{\kappa} (1 - e^{-2\kappa \Delta t_i}), \quad (24)$$

which is linear in $\mathbf{C}_i$. Together with the random-walk state equation

$$\mathbf{C}_{i+1} = \mathbf{C}_i + \boldsymbol{\eta}_i, \quad \boldsymbol{\eta}_i \sim \mathcal{N}(0, Q_i), \quad Q_i = 2D_c \Delta t_i, \quad (25)$$

this constitutes a standard state-space model. The two spatial coordinates are independent and share the same scalar variances, so the recursions below are applied to each coordinate separately.

A forward pass (Kalman filter) propagates the mean m_i and variance P_i of the center, conditioned on the observations up to $\mathbf{Y}_i$:

$$\begin{aligned} \text{predict: } m_i^- &= m_{i-1}, & P_i^- &= P_{i-1} + Q_{i-1}, \\ \text{update: } S_i &= H_i^2 P_i^- + R_i, & K_i &= \frac{H_i P_i^-}{S_i}, \\ m_i &= m_i^- + K_i (\mathbf{Y}_i - H_i m_i^-), & P_i &= (1 - K_i H_i) P_i^-, \end{aligned} \quad (26)$$

initialized with the Gaussian prior on $\mathbf{C}_0$. The innovations also yield the marginal log-likelihood of the data,

$$\log \pi(\mathbf{Y} | \theta) = \sum_i \log \mathcal{N}(\mathbf{Y}_i; H_i m_i^-, S_i), \quad (27)$$

where $\mathbf{C}$ has been integrated out analytically. A backward pass (Rauch–Tung–Striebel smoother) then conditions on the full trajectory to give the smoothed posterior mean,

$$J_i = \frac{P_i}{P_i + Q_i}, \quad m_i^s = m_i + J_i (m_{i+1}^s - m_i), \quad (28)$$

so that $m_i^s = \mathbb{E}[\mathbf{C}_i | \mathbf{X}, \theta]$ is the exact posterior-mean center at each frame.

The parameters κ and D_f are obtained by maximizing the marginal likelihood (Eq. 27) with D_c fixed; D_f may additionally be constrained to the independently measured slow diffusion coefficient. Conditional on the optimized parameters, a single filtering–smoothing pass returns the center path, which is used directly as the center estimate. The whole procedure is deterministic and scales linearly in the trajectory length.

Only long trajectories satisfying the focus-association criterion are used to infer the center trajectory. For shorter trajectories, the center displacement is assigned from the temporally nearest inferred center trajectory when the time difference is smaller than the threshold $\tau_{\max}$.

Note S2: Two foci models

To validate the subsampling method, we generate synthetic SPT data by following the description in Heltberg et al. (13).

The LLPS Model

We consider a particle diffusing in the cell nucleus. In the LLPS model, particle motion is described in three dimensions by the overdamped Langevin equation

$$d\mathbf{r} = dt \left[\nabla D(\mathbf{r}) - \frac{D(\mathbf{r})}{k_B T} \nabla U(\mathbf{r}) \right] + \sqrt{2D(\mathbf{r})} d\mathbf{W}, \quad (29)$$

where, $D(\mathbf{r})$ is the position-dependent diffusion coefficient, $U(\mathbf{r})$ is the potential associated with the focus, and $d\mathbf{W}$ a three-dimensional Wiener process. The term, $\nabla D(\mathbf{r})$, is the Itô drift associated with spatially varying diffusion. Together with the potential-dependent drift, it ensures that the equilibrium particle distribution follows the Boltzmann distribution.

We represent the focus by sigmoidal profiles of the diffusion coefficient and potential:

$$D(\mathbf{r}) = D_f + \frac{D_n - D_f}{1 + e^{-\beta(|\mathbf{r}| - r_f)}}, \quad U(\mathbf{r}) = \frac{v}{1 + e^{-\beta(|\mathbf{r}| - r_f)}}.$$

Here, D_f and D_n are the diffusion coefficients inside the focus and in the surrounding nucleus, respectively, r_f is the focus radius, v is the potential difference between the focus interior and the surrounding nucleus, and β controls the steepness of the transition at the focus boundary. With $v > 0$, the potential is lower inside the focus, making particle occupancy there energetically favorable. Parameter values are listed in Table S1. At equilibrium, the particle position follows the Boltzmann distribution

$$p(\mathbf{r}) = \frac{1}{Z} \exp \left[-\frac{U(\mathbf{r})}{k_B T} \right] \quad (30)$$

The PPPS Model

In the PPPS model, the focus contains N binding sites, each represented by a partially reactive sphere of radius r_b . The binding sites are confined to the focus by a potential $U_b(\mathbf{r})$, resulting in an equilibrium binding-site density $\rho(\mathbf{r}) \propto e^{-U_b(\mathbf{r})/k_B T}$. The position of each binding site evolves according to

$$d\mathbf{r} = -dt \frac{D_b}{k_B T} \nabla U_b(\mathbf{r}) + \sqrt{2D_b} d\mathbf{W}_t, \quad (31)$$

where D_b is the diffusion coefficient of a binding site. Because D_b is spatially constant, no Itô drift term is required. The potential, U_b , confines the binding sites but does not act directly on the particle.

An unbound particle diffuses with the constant diffusion coefficient D_n , both inside and outside the focus. Binding can occur when the particle encounters a binding-site boundary during a simulation time step dt . The binding probability is $p_b = \kappa \sqrt{\pi dt / D_n}$, where κ is the surface reaction rate. This reaction rule corresponds to the Robin boundary condition $D\mathbf{n} \cdot \nabla p(\mathbf{x}) = \kappa p(\mathbf{x})$ on the surface of a binding site, where $p(\mathbf{x})$ is the particle probability density at surface position $\mathbf{x}$, and $\mathbf{n}$ is the outward unit normal vector. While bound, the particle co-diffuses with the associated binding site and dissociates with rate k_- . Thus, in the PPPS model, particle enrichment within the focus arises from transient binding to mobile binding sites rather than from a direct potential acting on the particle.

Note S3: Deriving the mode of the density distribution as optimal observable

The local-density sampler returns a set $\{(\hat{\rho}_i, A_i)\}$ from N sampling disks placed uniformly in area inside the recording region, with $\hat{\rho}_i = X_i/A_i$ and X_i the particle count in disk i of area A_i . We derive the distribution of $\hat{\rho}_i$ under uniform-area sampling, show that its mode lies at the projected peak $\rho_{2D}(0)$, and characterise the finite-sample corrections. Throughout, $\rho(R)$ denotes the radial projected density on $R \in [0, r_f]$, assumed strictly monotone decreasing with $\rho(0) = \rho_{\max}$ and $\rho(r_f) = \rho_{\min}$.

Two distinct quantities must be kept separate in this derivation. The first is the density field $\rho(r)$, the number of localizations per unit area, which is maximal at the center, $\rho(0) = \rho_{\max}$. The second is the radial probability of sampling a localization at radius r : because the two-dimensional area element is $2\pi r dr$, this radial measure carries a Jacobian proportional to r and vanishes at the origin (below, $f_R(R) = 2R/r_f^2 \rightarrow 0$ as $R \rightarrow 0$). The vanishing at $r = 0$ is thus a property of the sampling measure, not of the density field. The estimator never evaluates the density at the single point $r = 0$; it returns the mode of the

distribution of local-density values $\{\hat{\rho}_i\}$, which, as shown below, equals the peak field value $\rho_{\max} = \rho(0)$.

Box centres $\mathbf{x} \in \mathcal{D}_f$ are drawn from $f_{\mathbf{x}}(x, y) = 1/(\pi r_f^2)$. To obtain the marginal distribution of $\rho = \rho(R)$, apply the polar Jacobian $dx dy = R dR d\theta$ followed by the scalar change of variable $R \rightarrow \rho$:

$$f_{R,\theta}(R, \theta) = \frac{R}{\pi r_f^2}, \quad f_{\rho,\theta}(\rho, \theta) = \frac{R(\rho)}{\pi r_f^2 |d\rho/dR|_{R=R(\rho)}}. \quad (32)$$

Integrating over θ (isotropy gives a factor 2π),

$$f_{\rho}(\rho) = \frac{2R(\rho)}{r_f^2 |d\rho/dR|_{R=R(\rho)}}. \quad (33)$$

The radial marginal $f_R(R) = 2R/r_f^2$ vanishes at $R = 0$: uniform-area sampling does not over-represent the centre. Any non-uniformity of f_{ρ} comes from the geometry of the map $\rho(R)$, not from the sampling.

In the limit $b \rightarrow \infty$ with $Z_L = r_f$, the projected profile is the chord $\rho(R) = 2\sqrt{r_f^2 - R^2}$. Differentiating gives $|d\rho/dR| = 4R/\rho$, and Eq. (33) reduces to

$$f_{\rho}(\rho) = \frac{\rho}{2r_f^2}, \quad \rho \in [0, 2r_f], \quad (34)$$

a normalised linear ramp. The mode of a monotone density is its right endpoint, so $\text{mode}(f_{\rho}) = 2r_f = \rho_{2D}(0)$.

This result determines how the focus-center correction affects the estimated partition free energy. The correction multiplies the density field by a Gaussian shrinkage, $\rho_c(r) = \rho(r) e^{-\alpha r^2}$ (main text). Because the partition free energy is read from the mode of the local-density distribution, and this mode sits at the peak of the field ($r = 0$), where $e^{-\alpha r^2} = 1$, the modal value $\rho_{\max}$ (and hereby also ΔU) is left unchanged by the correction. Equivalently, the map $\rho \mapsto \rho e^{-\alpha r(\rho)^2}$ leaves its right endpoint $\rho_{\max}$ (attained at $r = 0$) invariant, so $\text{mode}(f_{\rho_c}) = \text{mode}(f_{\rho}) = \rho_{\max}$. The radial Jacobian determines only how frequently sampling disks fall near the center; it does not enter the peak density that defines the mode. To consider the poissonian noise arising from finite-sample corrections, we consider the continuous f_{ρ} is observed through Poisson-noisy estimators $\hat{\rho}_i = X_i/A_i$ with $X_i \sim \text{Poisson}(\rho_i A_i)$. The empirical histogram is approximately f_{ρ} convolved with a per-disk Gaussian of relative width $1/\sqrt{\rho A}$.

Note S4: Investigating non-equilibrium currents in the data

The continuity equation for the probability density is $\partial_t p = -\nabla \cdot \mathbf{j}$, with probability current $\mathbf{j} = \mu p - D \nabla p$ (a drift term minus a diffusion term). Stationarity ($p = p(x)$) therefore requires $\nabla \cdot \mathbf{j} = 0$. Helmholtz-decomposing the current, $\mathbf{j} = \nabla \times (\psi \hat{z}) + \nabla \phi$, the divergence is carried entirely by the gradient part, $\nabla \cdot \mathbf{j} = \nabla^2 \phi$. Detailed balance can therefore fail in two independent ways that a two-dimensional position measurement can access:

- a source/sink current (divergence). This can affect the observed radial density and thereby also the observed potential.
- a rotational current (circulating). This does not affect the radial density, but does reveal a system out of equilibrium.

Divergence

A divergence current requires the ensemble to be open, i.e. where particles can be created and removed, expressed by the term, s : $\partial_t p = -\nabla \cdot \mathbf{j} + s$. To investigate this we measure where particles appear and disappear. If these are significantly different, we can reject the hypothesis of quasi-equilibrium. For each cell, we estimate the center by taking the median in x - and y -directions. Next, we take the first localization of every trajectory as its birth radius and the last as its death radius. To verify this measure we calculate the gap between first and last observations:

$$\delta = \langle r_{\text{birth}} \rangle - \langle r_{\text{death}} \rangle$$

The observed distributions, that is a result of gathering this for all cells, is now compared with the expected number that comes from our simulations in equilibrium. In this way we can also calibrate the method, as we can start the simulation out of equilibrium and measure the observed divergence.

Circulating current

To define the current we do as previously described in Battle et al., (*Science* 2016), where we aim to find a nonzero current is a model-independent signature of broken detailed balance. Defining in 2D:

$$a_i = (\mathbf{r}_i \times \mathbf{r}_{i+1}) = x_i y_{i+1} - x_{i+1} y_i$$

We numerically assign the midpoint $\bar{x}_i = \frac{1}{2}(x_i + x_{i+1})$ and the jump $\Delta x_i = x_{i+1} - x_i$ so $a_i = \bar{x}_i \Delta y_i - \bar{y}_i \Delta x_i$. Now the current and the related angular drift can be estimated respectively as:

$$J = \langle a_i \rangle / (2\Delta t) \quad \text{and} \quad \Omega = \langle a_i \rangle / (2\Delta t \langle \bar{r}^2 \rangle)$$

Testing against simulation

All tests are calibrated against the same condensate model, driven in the specific way each test is meant to detect. The focus is a sigmoidal well of radius r_f and depth A ,

$$\frac{U(r)}{k_B T} = \frac{A/k_B T}{1 + e^{-\beta(r-r_f)}},$$

and a molecule follows the overdamped Langevin equation:

$$d\mathbf{r} = -\frac{D}{k_B T} \nabla U dt + \sqrt{2D} d\mathbf{W}.$$

Simulating this, with initial positions following the Boltzmann distribution $p \propto e^{-U/k_B T}$, we tested this for the equilibrium system first. Next we added a non-equilibrium control in the following way:

- **Outward divergence.** We assume a fraction of particles are "pumped" into the system, so their initial position is between $[r_f/4 : r_f]$, and we absorb them if they cross a boundary $r > 0.9 * r_f$. This therefore gives an outward flux.
- **Inward divergence.** Instead of starting the particles in their equilibrium distribution, we start them with a random probability everywhere. This means that we effectively measure an enhancement of density inside the condensate and therefore an inward divergence.
- **Curl force.** We add a non-conservative rotational drift:

$$d\mathbf{r} = \left[-\frac{D}{k_B T} \nabla U + \Omega_0 \hat{z} \times \mathbf{r} \right] dt + \sqrt{2D} d\mathbf{W}, \quad \text{where} \quad \hat{z} \times \mathbf{r} = (-y, x),$$

whose nonzero curl sustains a circulating current of rate Ω_0 .

Like in the main text, we add Gaussian localization noise ($\sigma = 25$ nm). On Fig. S3A, we observe representative cells in equilibrium, and out of equilibrium. For the radial current, we calculate δ in the equilibrium system, which we find to be approximately zero, and find the 95% confidence interval to compare with systems where we add currents (Fig. S3B). Using this measure for the radially incuded currents, we compare the equilibrium results (blue) to the non-equilibrium simulations (green and red), revealing that the mean is significantly outside the 95% confidence interval for both tests (Fig. S3C). Doing the same for the rotaional current, we once again start by finding the equilibrium simulation to give approximately zero (Fig. S3D), whereas the current is clearly detected in the non-equilibrium simulation (Fig. S3E). These results show that our methods do detect broken detailed balance, whereas they agree with zero in the equilibrium case as we would expect.

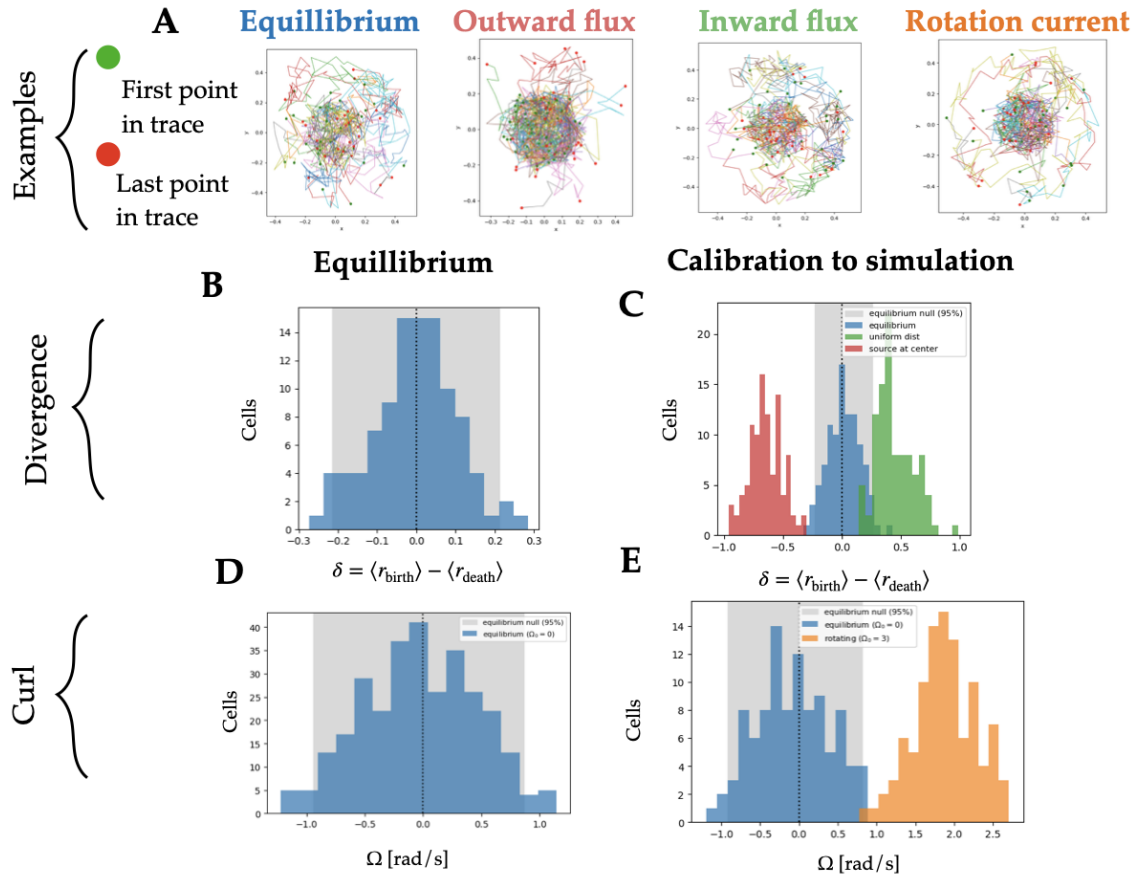


Figure S1: A) Example traces for equilibrium, inward divergence, outward divergence and rotational currents. B) Measuring the gap δ in equilibrium. Grey line defines the 95% confidence interval. C) Measurements of δ for the two systems with radial current. D) Same as B, but for the curl of each trace. E) Measurements of curl for cells, in the simulated system with inserted rotational current.

Variable	Model	Description	Value	Range	Units
r_f	All	Radius of the focus	0.1	0.05-0.2	μm
r_n	All	Radius of the nucleus	0.5	0.3-1	μm
D_n	All	Diffusion coefficient in nucleus	1.0	0.5 - 2.0	$\mu\text{m}^2/\text{s}$
D_c	All	Diffusion coefficient of the focus center	0.005	0.001-1	$\mu\text{m}^2/\text{s}$
D_f	LLPS	Diffusion coefficient in the focus	0.05	0.01-0.5	$\mu\text{m}^2/\text{s}$
D_b	PPPS	Diffusion coefficient of binding sites	0.005	0-0.1	$\mu\text{m}^2/\text{s}$
σ	All	Experimental localization noise	0.03	-	μm
v	LLPS, Sigmoid	Surface potential	5.0	0-10	$k_B T$
β	LLPS, Sigmoid	Steepness in potential	1000	-	μm^{-1}
ρ	PPPS	Density of binding sites	2E4	2.8E4 - 6.8E4	μm^{-3}
r_b	PPPS	Radius of binding site potential	0.1	0.05-0.2	μm
k_-	PPPS	Unbinding rate	500	0 - 1000	s^{-1}
k_A	PPPS	Absorption parameter	100	0-300	$\mu\text{m}/\text{s}$

Table S1: Parameters used in the various models described throughout this work.

Note S5: Two state diffusion validation

We investigate the number of diffusive states considered over the global fit for each cell type. To quantify this we fitted $K = 1, 2$, and 3-component Rayleigh mixtures by maximum likelihood to the pooled displacement lengths.

We calculate the CDF of each distribution and quantify each fit by its Kolmogorov-Smirnoff distance, the largest deviation between the empirical and fitted CDF. The one-component model fails badly, $KS = 0.20\text{--}0.34$ across the datasets. The two-component fit reduces this roughly tenfold, to $KS = 0.021\text{--}0.042$. We note that this is comparable to the finite-sample fluctuation level expected if the distribution was true. This is given as: $1.36/\sqrt{n} \approx 0.02$ (where 1.36 is the 95% quantile of the scaled KS statistic) for our sample sizes with $n \approx 5 \cdot 10^3$. This means that the two-component residual is about as small as it is statistically expected to be. The three-component fit improves the distance only marginally further, to 0.005–0.008, but as this is below the sampling-noise level, we might be fitting the noise rather than actual diffusivity.

Note S6: Robustness to the focal depth and the anchor-trace threshold

Two parameters of the analysis pipeline are set externally rather than fitted: the half-depth of the observable focal slab, Z_L , and the minimal length of the anchor trajectories used for the focus-center inference, $N_{\min}$. Here we quantify how sensitive the estimated partition free energy is to both choices.

Focal depth Z_L

The focal depth is a property of the imaging system (the depth of field of the objective), and it enters the analysis only through the projected target ΔU_{2D} , i.e. through the axial integration of the three-dimensional density over the slab $|z| < Z_L$. In the experimental setup, the camera Andor EMCCD, was used which has a calibrated focal plane of $Z_L = 150 \text{ nm}$ (14). To quantify the consequence of an imprecise value, we simulate trajectories in a focal slab of half-depth $Z_L = 0.15 \mu\text{m}$ and compare the recovered potential difference with the theoretically projected well depth evaluated over a plausible range of focal depths (Fig. S3). The estimate agrees with the projected depth at the true Z_L within the standard error. Because a deeper slab admixes more out-of-focus background, the projected well depth decreases slowly with Z_L (from $4.5 k_B T$ at $0.15 \mu\text{m}$ to $4.0 k_B T$ at $0.25 \mu\text{m}$), so the relative error incurred by assuming $Z_L = 0.15 \mu\text{m}$ grows approximately linearly with the misestimation and remains below 13% ($\lesssim 0.5 k_B T$) even for an error as large as $0.1 \mu\text{m}$. Since the same Z_L applies to all datasets, such an offset would furthermore shift all conditions coherently in the diffusivity–density plane rather than alter their relative placement, leaving the LLPS/PPPS classification unaffected.

Anchor-trace threshold $N_{\min}$

The threshold $N_{\min}$ selects which trajectories serve as anchors for inferring the focus-center path; it is not a filter on the localizations entering the density estimate, since all recorded positions are subsequently transformed into the focus frame. To

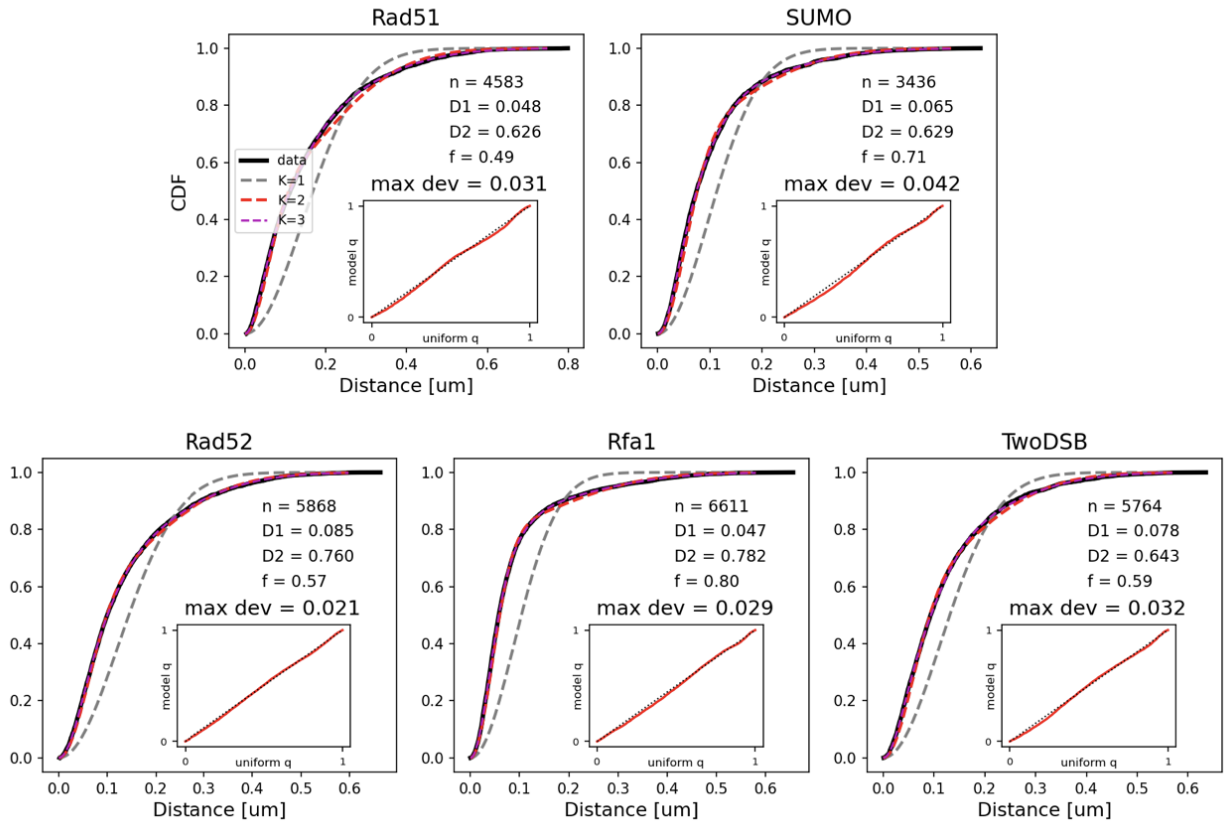


Figure S2: Calculated CDF for all cell types. This is compared to the 1,2 and 3 mixture Raleigh models. Inset represents the QQ-plot, revealing the maximal deviation between the CDF and the 2-state model.

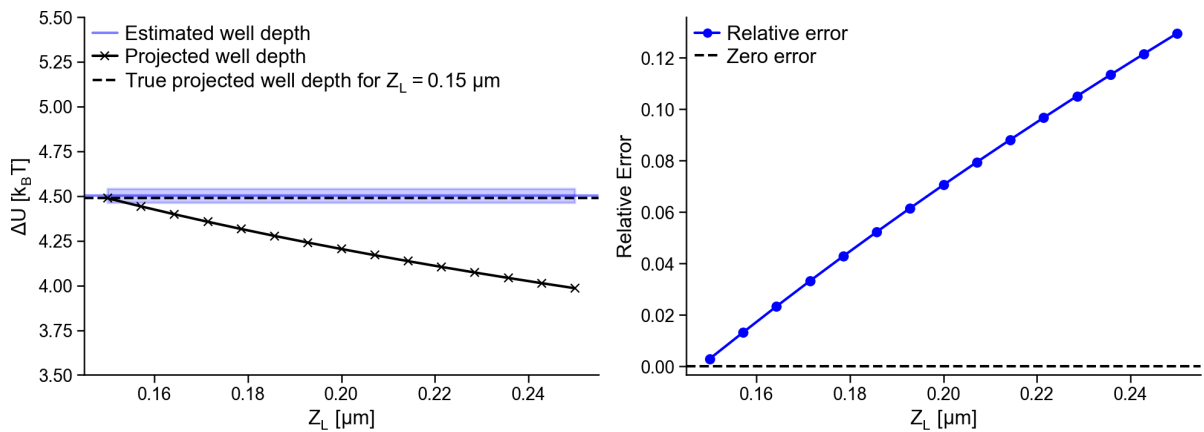


Figure S3: Estimated ΔU for simulated data contained to a focal slab of half-depth $Z_L = 0.15 \mu\text{m}$ is compared to the theoretical projected well depth as a function of Z_L . The error bands show the standard error on the estimated potential difference. The right-hand-side shows the corresponding relative error between the estimated $\Delta U(Z_L = 0.15 \mu\text{m})$ and the theoretical value for varying Z_L .

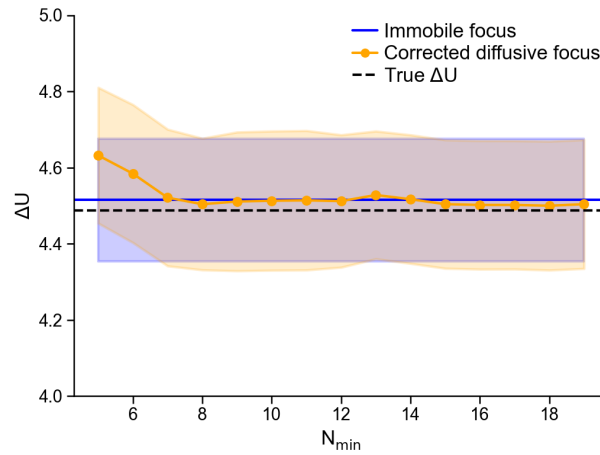


Figure S4: Estimated ΔU as a function of the length of anchor traces, $N_{\min}$. Error bands show the standard error on the estimated potential difference.

test for a selection bias toward long-lived molecules, we simulate a diffusing focus with a known potential difference and recover ΔU for $N_{\min} = 5$ –19 (Fig. S4). The corrected estimate is indistinguishable from both the immobile-focus reference and the true value over the whole range, with only a small upward deviation for the shortest anchors ($N_{\min} \lesssim 6$), where fast nucleoplasmic trajectories can occasionally be misassigned as anchors. The working value $N_{\min} = 10$ lies well inside this plateau, and the estimated ΔU is not vulnerable to this precise choice.